Julià Delos Ayllón

Hardware and Software Engineer · Failed Entrepreneur · Linux Nerd · From Embedded to Full-Stack

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Profile

Seasoned engineer and failed entrepreneur with a PhD in Microelectronics, bringing together a deep understanding of hardware and software and a passion for building things. Enjoys turning theory into working systems and uses Blender to bring personal engineering and 3D printing projects to life. Eager to contribute this experience, curiosity, and Mediterranean passion to Blender's development.

Technical skills

Languages: C, C++, TypeScript, JavaScript, Python, Java, Solidity, MATLAB

Systems: Linux, embedded Linux, firmware, ARM, real-time systems (dSPACE, RTAI), CAN, edge computing, WebAssembly

Frameworks: React, Next.js, Remix, Astro, Gatsby, Django, Hardhat, Redux, XState, Prisma, Sanity, Shopify apps

Tooling: Git, ClearCase, GitHub Actions, Docker, Fly.io, Vitest, Playwright, Cypress, Mastra

IDEs: LazyVim (Neovim), VS Code

Hardware: Power electronics, circuit design, MATLAB/Simulink, prototyping and test setups

Experience

La Calçotada, Founder – Netherlands 2019 – 2025

- Turned a yearly food festival into a food-experience business serving up to **10,000 users** across the Netherlands and seven neighbouring countries, building the e-commerce flow, logistics and supplier network.
- Generated **EUR 100k+** in online sales during the initial market validation phase, reaching a peak run-rate of around **EUR 20k/month**, with gross margins above **40%** and approximately **3%** e-commerce conversion.
- Partnered with restaurants in Amsterdam, Eindhoven and Tilburg to host around **200 guests** per afternoon during otherwise quiet or closed hours, generating approximately **EUR 19 per guest** for each restaurant.
- Built an equity-crowdfunding platform intended to give the community ownership of the business, applying open-source principles to an open business model; the initiative failed at the go-to-market stage.

Ripatrip, Co-Founder and CTO – Remote (team in Portugal) Sept 2019 – June 2020

- Co-founded a travel search platform supported by **Startup Lisboa**, initially focused on kitesurfing and later expanded to ski and snowboard destinations.
- Delivered a working demo within **two weeks** and presented it at **Web Summit 2019 in Lisbon**. Guided **two junior developers** through further development.
- Enabled travel searches across **5,000 destinations** by integrating multiple APIs for destination data and travel offerings.
- Built the platform with React-Gatsby/Redux and a CMS as the only backend, querying flight, accommodation, and destination data directly from the client in real time.

ICT Group, Innovation and Technology Consultant – Eindhoven, Netherlands Sept 2016 – Sept 2018

- Worked closely with the C-suite to introduce visual business modelling tools across business units, creating a shared language for strategy and innovation.
- Convinced senior leadership to establish a **dedicated data-science team and data lake** by showing business-unit leaders how data science could benefit their units.
- Coached around **10 developer-led teams** for the internal Dragons' Den, using experiments to validate technical ideas and customer value.

Philips Research, Research Scientist – Eindhoven, Netherlands June 2011 – Dec 2015

- Surpassed state-of-the-art integrated power density by developing a new power-system-on-chip LED driver.
- Enabled chip-area optimization by creating a design framework based on a closed-form model, validated theoretically and experimentally.
- Expanded Philips' patent portfolio through circuit inventions resulting in **8 first-named EPO patent records (6 granted)**.
- Completed a PhD with Eindhoven University of Technology, publishing peer-reviewed research and presenting internationally.

Lear Corporation, Hardware Engineer – Valls, Spain Mar 2010 – May 2011

- Delivered a **60 kW** racing-car inverter that worked on its first iteration by leading hardware design, test setup, and production.
- Met an inverter volume target considered infeasible by coordinating suppliers and technical contributors.
- Reduced current-sensor cost to **one-sixth** by designing an in-house, low-profile tangential sensor with adjustable sensitivity and high accuracy.

HP (via Capgemini), Embedded Systems Engineer, Printer Firmware – Barcelona Sept 2006 – Sept 2007

- Delivered production fixes within the **first month** in a large C/C++ and Java codebase spanning Linux and distributed ARM targets, with a **six-hour full build**.
- Resolved longstanding firmware issues by applying hardware expertise beyond the software team's capabilities.
- Managed builds, release packaging, and security updates for HP's top-of-the-line large-format printers using ClearCase.
- Transferred firmware knowledge to the Bangalore team through **two-to-three weeks** of hands-on training.

Education

PhD	**Eindhoven University of Technology**, PhD in Electrical Engineering	2016
MSc	**Technical University of Catalonia**, MSc in Electronics Engineering	2010
BSc	**Polytech Lille**, BSc in Industrial Software Development	2006
BSc	**Technical University of Catalonia**, BSc in Electrical Engineering	2005

Languages

Native Catalan and Spanish · Proficient in English and French · Conversational in Portuguese and Dutch

Projects

Switched-capacitor converter impedance tools

- Porting my MATLAB converter models to C++ (SymEngine) and **WebAssembly** GitHub Actions builds the pinned library and checks the native and WebAssembly results against MATLAB.

Onion Memorandum platform July 2024 – Mar 2025

- Remix (Epic Stack) application with Prisma migrations, MDX content and PDF generation, tested with Vitest and Playwright and deployed with Docker on Fly.io via GitHub Actions.

QR ticketing app and ticket reader Nov 2023 – Mar 2024

- Shopify app that issues QR-code tickets as PDFs, plus a Remix scanner app built on XState; tested with Cypress and Vitest, deployed with Docker on Fly.io.

Aircraft power converter, Polytechnic Institute of New York University Feb 2009 – Dec 2009

- As a research visitor, designed and modelled a **20 kW** fuel-cell converter providing ±270 V rails for a universal power-converter research system.

Embedded and real-time internships, Renault Technocentre, LAGIS and INRETS Feb 2005 – July 2006

- Benchmarked vehicle-speed estimators on dSPACE real-time hardware with MATLAB/Simulink (Renault).
- Wrote a CAN network sniffer in C on Linux/RTAI for the Robucar vehicle and the developer manual for SynDEx, a tool for distributed real-time embedded systems (LAGIS).
- Designed an electronic system to detect electromagnetic perturbations in antenna EMC test setups (INRETS).